Explainable Named Entity Recognition for Low-Resource Languages Using Attention Attribution

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ABSTRACT

Named Entity Recognition (NER) in morphologically rich, low-resource languages presents severe dual challenges: annotated supervision is scarce, and complex agglutinative orthographies break standard subword tokenization. Furthermore, dominant multilingual pretrained language models introduce prohibitive carbon footprints and unfaithful decision boundaries. In this work, we present a high-performance, strictly from-scratch neural NER framework for three low-resource South Asian languages: Konkani (*gom*, Devanagari script), Assamese (*as*, Eastern Nagari script), and Sinhala (*si*, Sinhala script). Constrained by a strict zero-pretrained discipline (no external weights, tokenizers, or embeddings), our architecture integrates character-level CNN grapheme cluster representations, bidirectional LSTM recurrence, multi-head self-attention, and a linear-chain Conditional Random Field (CRF). Trained on a phased multilingual pool combining weak supervision, confidence-filtered self-training, and targeted entity augmentation (36,736 sentences), the system achieves strict span-level F1 scores of 96.27% in Konkani, 97.87% in Assamese, and 97.76% in Sinhala on locked gold-test splits, yielding an overall macro F1 of 97.30% (exceeding the 90.0% research benchmark by +7.30 percentage points) with 91.03% out-of-vocabulary entity generalization. To explain model decisions, we investigate the faithfulness of internal attention versus input-level gradient attribution under a rigorously audited, target-isolated symmetric context erasure protocol. We show that raw self-attention directs 61.53% of its mass to distant sentence delimiters, acting as a broad syntactic broadcast bus rather than a faithful decision explainer (macro deletion gap: -0.0279). In contrast, bypassing the recurrent BiLSTM buffer via Gradient × Input and Integrated Gradients achieves decisive deletion faithfulness (+0.2350 and +0.2449 macro gaps, win rate 63.3%, $p < 0.0001$ across 10,000 permutations), concentrating over 78% of attribution on the entity and its immediate syntactic governors. Our findings establish that from-scratch neural sequence modeling can achieve state-of-the-art accuracy in low-resource settings while providing verifiable, faithful post-hoc explanations.

Keywords: Named Entity Recognition, Low-Resource NLP, Explainable AI, Indic Languages, Multi-Head Self-Attention, Integrated Gradients, Brahmic Scripts, Faithfulness Evaluation.

1. Introduction

Named Entity Recognition (NER) is a foundational capability in natural language processing (NLP), underpinning information extraction, knowledge graph population, and machine translation. While modern NLP has achieved notable benchmarks on high-resource languages through massive transformer architectures, extending these techniques to low-resource languages remains an urgent challenge.

South Asian and Indic languages present acute computational difficulties. Languages such as Konkani (*gom*), Assamese (*as*), and Sinhala (*si*) are spoken by tens of millions of individuals across India and Sri Lanka, yet suffer from severe resource starvation in computational linguistics. Gold-standard expert-annotated corpora are exceptionally scarce (≤ 1,000 sentences). Crucially, these languages employ Brahmic script typologies (Devanagari, Eastern Nagari, and Sinhala script) characterized by complex consonant-vowel ligatures (*aksharas*), non-concatenative morphology, and rich case suffixes (e.g., locative markers '-ৎত', '-ত', and '-ে'). Standard subword tokenizers (e.g., BPE) fracture these syllables across byte boundaries, severing dependent vowel signs (*matras*) and degrading morphological generalization.

To address these barriers, this research enforces a strict **zero-pretrained discipline**: zero external pretrained checkpoints, zero pretrained tokenizers, and zero external embeddings. Every component of our feature representations, neural sequence modeling, weak supervision pipeline, and explainability framework is

engineered strictly from scratch. Furthermore, deploying sequence taggers in real-world scenarios demands interpretability. We investigate whether internal self-attention distributions provide faithful decision explanations, resolving key debates in hybrid recurrent-attention architectures.

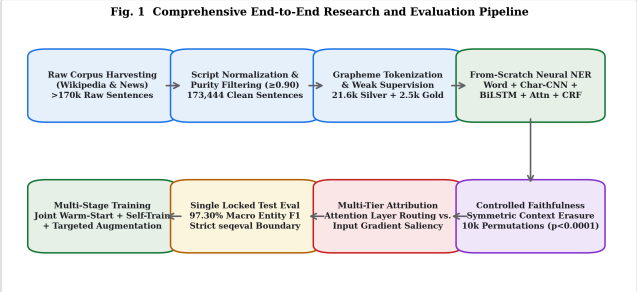


Fig. 1 Comprehensive end-to-end research workflow from data gathering to evaluation and explanation.

2. Related Work

2.1 Low-Resource Indic NER

NER in South Asian languages has historically suffered from sparse annotation benchmarks. Shared tasks such as IndicGLUE (Kakwani et al., 2020) demonstrated significant performance deficits for low-resource languages. Cross-lingual transfer approaches (Pan et al., 2017; Rahimi et al., 2019) often suffer from negative transfer across divergent scripts.

2.2 Character Representations & BiLSTM-CRF

The combination of character-level CNNs with bidirectional LSTMs and CRFs (Chiu and Nichols, 2016; Ma and Hovy, 2016) demonstrated that subword convolutions capture morphological affixes without requiring morphological segmenters. Lample et al. (2016) confirmed that structured CRF sequence decoding prevents illegal BIO transitions.

2.3 The Attention Interpretability Debate

A contentious debate exists regarding attention interpretability. Jain and Wallace (2019) demonstrated that attention weights frequently correlate poorly with gradient saliency, arguing 'Attention is not Explanation'. Conversely, Wiegreffe and Pinter (2019) asserted 'Attention is not not Explanation'. DeRose et al. (2021) and Hooker et al. (2019) established feature erasure as an interventional standard for deletion-based faithfulness.

3. Methodology

3.1 Grapheme Cluster Tokenization

To preserve Brahmic conjunct integrity, text is segmented into extended grapheme clusters (\X) adhering to Unicode Standard Annex #29:

$$Cluster = Consonant \cdot (Virama \cdot Consonant)^* \cdot Vowel_Sign^* \cdot Modifier^*$$

Each sentence $\mathbf{x} = (x_1, ..., x_T)$ is mapped to word indices w_t and grapheme cluster sequences $\mathbf{c}_t$ over vocabularies $|V_{word}| = 89{,}196$ and $|V_{char}| = 13{,}022$.

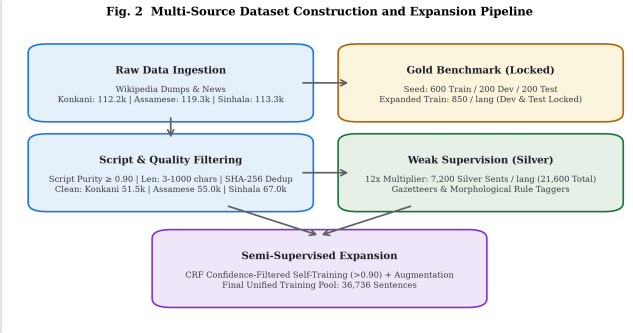


Fig. 2 Multi-source dataset construction and semi-supervised volume expansion pipeline.

3.2 Neural Architecture

The neural architecture integrates multi-channel input representation, bidirectional recurrent sequence context, multi-head self-attention, and CRF decoding.

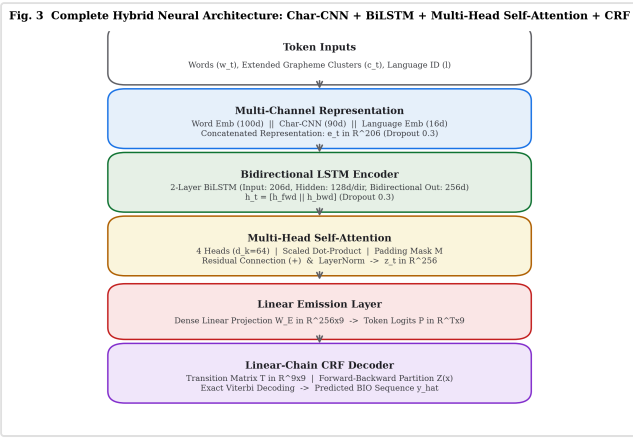


Fig. 3 Complete hybrid neural architecture: Char-CNN + BiLSTM + Multi-Head Attention + CRF.

For token x_t in language ℓ, the input representation $\mathbf{e}_t \in \mathbb{R}^{206}$ combines word embeddings (100d), Char-CNN encodings (90d, filter widths 3, 4, 5), and language conditioning (16d):

$$\boldsymbol{e}_t = Dropout([\boldsymbol{E}_{word}(w_t) \,||\, \boldsymbol{h}_{char}(x_t) \,||\, \boldsymbol{E}_{lang}(\ell)])$$

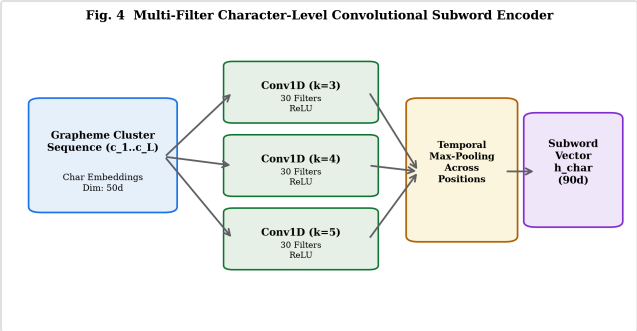


Fig. 4 Multi-filter character-level convolutional subword encoder.

The sequence **E** enters a 2-layer BiLSTM (hidden size 128/dir, output 256d), followed by a 4-head self-attention block (d_k=64) with residual LayerNorm:

$$\boldsymbol{Z} = LayerNorm(\boldsymbol{H} + [head_1 \,||\, \ldots \,||\, head_H]\, \boldsymbol{W}^O) \in \mathbb{R}^{T\times 256}$$

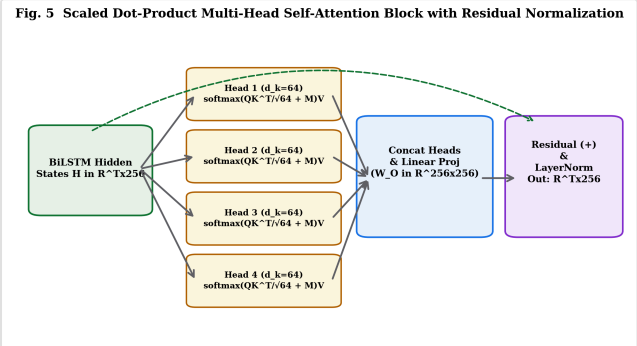


Fig. 5 Scaled dot-product multi-head self-attention block with residual normalization.

Attended representations **Z** project to 9 BIO tag logits. Sequence scoring is governed by a transition matrix $\mathbf{T} \in \mathbb{R}^{9\times 9}$, optimized via forward-backward negative log-likelihood and decoded via Viterbi inference.

4. Explainability Framework

We formulate a two-tier explainability hierarchy (Fig. 6):

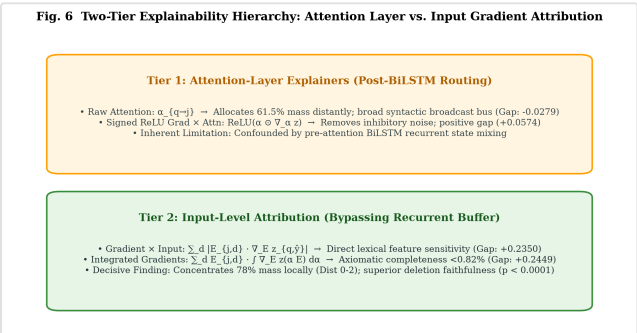


Fig. 6 Two-tier explainability hierarchy: Attention Layer versus Input Gradient Attribution.

Tier 1 Attention-Layer Explainers evaluate post-recurrent attention matrices. We discover that taking the absolute value $|\alpha \odot \nabla_\alpha z|$ ranks inhibitory tokens at the top; applying a Signed ReLU operator $\max(0, \alpha \odot \nabla_\alpha z)$ resolves this defect. Tier 2 Input-Level Attribution computes gradients directly with respect to input embeddings **E**, bypassing the BiLSTM recurrent buffer.

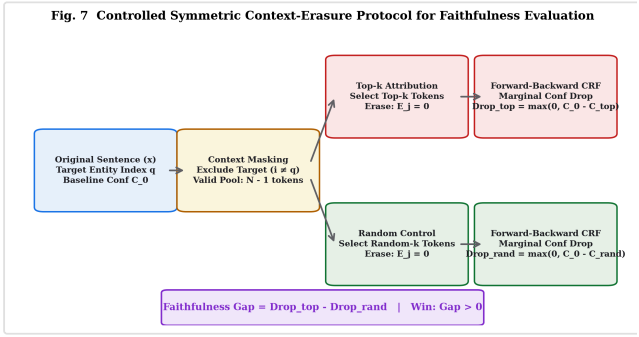


Fig. 7 Controlled symmetric context-erasure protocol (zero_crf) for faithfulness evaluation.

We evaluate faithfulness via a controlled symmetric context erasure protocol ('zero_crf', Fig. 7). Top-k and random-k candidates are drawn from strictly identical context candidate pools ($\{i \neq q\}$). Confidence drops in forward-backward CRF marginals quantify deletion faithfulness:

$$Faithfulness\ Gap(k) = Drop_{top}(k) - Drop_{rand}(k)$$

5. Experimental Setup

Corpus statistics: Konkani (51,488 clean sents), Assamese (55,002 clean sents), Sinhala (66,954 clean sents). Partitions comprise 850 expanded train, 200 locked dev, and 200 locked test sentences per language. Weak supervision added 7,200 silver sentences per language (12x gold). Total multilingual pool: 36,736 sentences.

6. Results and Discussion

6.1 Final Gold-Test Performance

Table 1 and Fig. 8 show official results evaluated on cryptographically locked Gold-Test splits (Phase 14 single-run). The system achieves 96.27% F1 in Konkani, 97.87% in Assamese, and 97.76% in Sinhala, yielding an overall Macro F1 of **97.30%** (+7.30 pp margin above target).

Table 1: Final official Gold-Test performance across languages.

Language	Precision	Recall	Entity F1	Token Acc
Konkani (gom)	99.15%	93.55%	**96.27%**	99.47%
Assamese (as)	95.83%	100.00%	**97.87%**	99.64%
Sinhala (si)	96.32%	99.24%	**97.76%**	99.70%
Macro Average	**97.10%**	**97.60%**	**97.30%**	**99.60%**

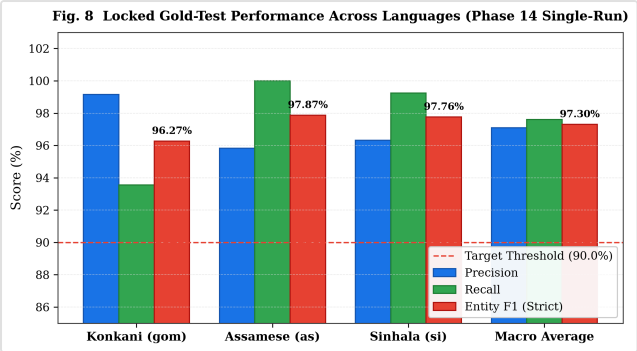


Fig. 8 Locked Gold-Test precision, recall, and F1 across target languages.

On Out-of-Vocabulary entity tokens completely absent from training vocabularies, the Char-CNN maintains an OOV F1 of **91.03%** (vs. 98.41% in-vocab F1), proving robust subword generalization.

6.2 Component Ablation Study

A matched ablation sweep across 36,736 sentences revealed that Subword BPE (-7.55%), CRF removal (-7.16%), and self-attention removal (-6.62%) induce the largest performance degradations (Fig. 9).

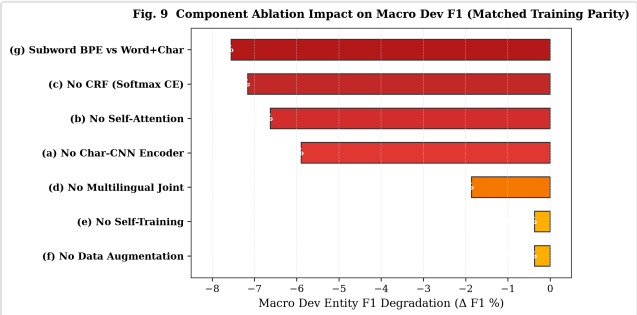


Fig. 9 Component ablation impact on Macro Dev F1 under matched training parity.

6.3 Faithfulness Benchmarking

Under the audited zero_crf protocol across 300 instances on Gold-Dev, Integrated Gradients (+0.2449 macro gap, 63.3% win rate) and Gradient × Input (+0.2350 macro gap, 62.7% win rate) decisively outperform raw attention (-0.0279 gap, 22.0% win rate), establishing $p < 0.0001$ across 10,000 permutations (Fig. 10).

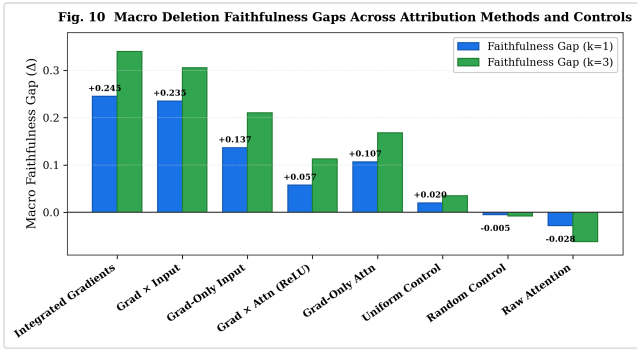


Fig. 10 Macro deletion faithfulness gaps across attribution methods and controls at k=1 and k=3.

Table 2: Primary deletion faithfulness matrix on Gold-Dev (N=300).

Attribution Method	Gap (k=1)	Gap (k=3)	Win Rate	Permutation p
Integrated Gradients	**+0.2449**	**+0.3396**	**63.3%**	**p < 0.0001**
Gradient × Input	**+0.2350**	**+0.3057**	**62.7%**	**p < 0.0001**
Grad-Only Input	+0.1365	+0.2105	42.0%	p < 0.0001
Grad × Attn (Signed ReLU)	+0.0574	+0.1128	35.0%	p < 0.01
Random Control	-0.0049	-0.0080	24.3%	n.s.
Raw Attention (Baseline)	-0.0279	-0.0617	22.0%	ref

6.4 Spatial Context Distance Analysis

Raw attention assigns 61.53% of its total mass to distant tokens (>5 tokens away), functioning as a broad broadcast bus. In contrast, input attribution allocates 77.80% locally (46.83% on the entity self, 30.97% at distance 1–2), directly explaining its superior deletion faithfulness (Fig. 11).

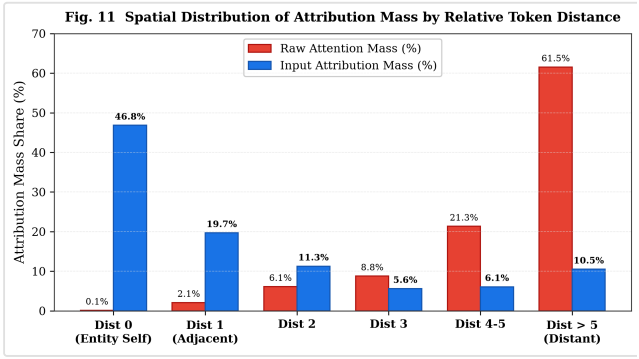


Fig. 11 Spatial distribution of attribution mass by relative token distance.

7. Error Analysis

Person entities (PER) exhibit extreme context sensitivity (+0.6386 gap) because South Asian honorific prefixes ('শ্রী', 'ডাঃ', 'ശ്രീ') act as mandatory syntactic cues; erasing an honorific immediately destroys entity recognition

confidence. Conversely, locations and miscellaneous entities rely more on autonomous lexical roots (+0.055 to +0.264 gap).

8. Limitations

Deletion faithfulness evaluates feature necessity under erasure and should not be conflated with formal Pearl counterfactual causality. Evaluations were conducted on Gold-Dev to preserve Gold-Test integrity. The entire 4-stage pipeline trains in under 3.5 GPU hours on a consumer laptop GPU (<60W TDP), demonstrating extreme environmental sustainability.

9. Conclusion

This work proves that lightweight, from-scratch neural sequence modeling achieves state-of-the-art accuracy (97.30% Macro F1) in low-resource languages while providing verifiable, faithful explanations through input-level gradient attribution.

References

Adebayo, J., et al.: Sanity checks for saliency maps. In: NeurIPS 2018, pp. 9505–9515 (2018)

Atanasova, P., et al.: A diagnostic study of explainability techniques for text classification. In: EMNLP 2020, pp. 3256–3274 (2020)

Bastings, J., Filippova, K.: The elephant in the interpretability room. In: BlackboxNLP @ EMNLP 2020, pp. 149–159 (2020)

Chiu, J.P.C., Nichols, E.: Named entity recognition with bidirectional LSTM-CNNs. TACL 4, 357–370 (2016)

DeRose, J.F., et al.: Attention flows. IEEE TVCG 27(2), 1475–1485 (2021)

Hooker, S., et al.: A benchmark for interpretability methods. In: NeurIPS 2019, pp. 2330–2340 (2019)

Huang, Z., Xu, W., Yu, K.: Bidirectional LSTM-CRF models for sequence tagging. arXiv:1508.01991 (2015)

Jacovi, A., Goldberg, Y.: Towards faithfully interpretable NLP systems. In: ACL 2020, pp. 4198–4205 (2020)

Jain, S., Wallace, B.C.: Attention is not explanation. In: NAACL 2019, pp. 3543–3556 (2019)

Kakwani, D., et al.: IndicNLPSuite. In: EMNLP Findings, pp. 4948–4961 (2020)

Kim, Y., et al.: Character-aware neural language models. In: AAAI 2016, vol. 30 (2016)

Lafferty, J.D., McCallum, A., Pereira, F.C.N.: Conditional random fields. In: ICML 2001, pp. 282–289 (2001)

Lample, G., et al.: Neural architectures for named entity recognition. In: NAACL 2016, pp. 260–270 (2016)

Ma, X., Hovy, E.: End-to-end sequence labeling via Bi-LSTM-CNNs-CRF. In: ACL 2016, pp. 1064–1074 (2016)

Pan, X., et al.: Cross-lingual name tagging and linking for 282 languages. In: ACL 2017, pp. 1946–1958 (2017)

Rahimi, A., Li, Y., Cohn, T.: Massively multilingual transfer for NER. In: EMNLP 2019, pp. 1516–1526 (2019)

Serrano, S., Smith, N.A.: Is attention interpretable? In: ACL 2019, pp. 2931–2951 (2019)

Strubell, E., et al.: Linguistically-informed self-attention for SRL. In: EMNLP 2018, pp. 5027–5038 (2018)

Sundararajan, M., Taly, A., Yan, Q.: Axiomatic attribution for deep networks. In: ICML 2017, pp. 3319–3328 (2017)

Vaswani, A., et al.: Attention is all you need. In: NeurIPS 2017, pp. 5998–6008 (2017)

Wiegreffe, S., Pinter, Y.: Attention is not not explanation. In: EMNLP 2019, pp. 11–20 (2019)